SS-C-621b
January 19, 1968
SUPERSEDING
Int. Fed. Spec. SS-C-00621a(GSA-FSS)
April 18, 1966 and
Fed. Spec. SS-C-621
April 28, 1931

FEDERAL SPECIFICATION

CONCRETE MASONRY UNITS, HOLLOW (AND SOLID, PREFACED AND UNGLAZED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers concrete masonry units for interior or exterior, protected or unprotected walls, above or below grade.

1.2 Classification. Concrete masonry units furnished under this specification shall be of the following types, classes, grades, forms, and styles, as specified (see 6.2):

Types:
- Type I - Moisture controlled.
- II - Nonmoisture controlled.

Classes and grades:
- Class 1 - Hollow, load bearing.
  - Grade G - Protected walls above grade.
  - P - Protected exterior walls above and below grade.
  - U - Unprotected exterior walls above and below grade.
- Class 2 - Solid, load bearing.
  - Grade G - Protected walls above grade.
  - P - Protected exterior walls above and below grade.
  - U - Unprotected exterior walls above and below grade.
- Class 3 - Hollow, nonload bearing.

Forms:
- Form A - Unglazed.
- B - Prefaced (type I, grade U and type I, class 3 only).

Styles:
- Style a - Regular (line block).
- b - Corner.
- c - Double corner (header block).
- d - Jamb (slots).
- e - Termite.
- f - Bull nose.
- g - Bond seam.
- h - Beam lintel.
- i - Control joint.
- j - Special and other shapes.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

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Federal Specifications:

TT-T-295 - Thinner, Paint, Mineral Spirits, Volatile, "Odorless".
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads (40" x 48" 4-Way Partial and 4-Way Pallets).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publications:

C 90 - Specification for Concrete Masonry Units, Hollow-Load-Bearing.
C 129 - Specification for Concrete Masonry Units, Hollow, Non-Load Bearing.
C 140 - Methods of Sampling and Testing Concrete Masonry Units.
C 145 - Specification for Concrete Masonry Units, Solid Load-Bearing.
C 426 - Method of Test for Drying Shrinkage of Concrete Block.
E 84 - Method of Test for Surface Burning Characteristics of Building Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

American Trucking Associations, Inc.:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

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Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1.1 Types I and II, class 1. Hollow load bearing units shall meet the requirements in ASTM C 90 (see 4.4.1).

3.1.2 Types I and II, class 2. Solid load bearing units shall meet the requirements in ASTM C 145 (see 4.4.1).

3.1.3 Types I and II, class 3. Hollow nonload-bearing units shall meet the requirements in ASTM C 129 (see 4.4.1).

3.2 Form B, prefaced.

3.2.1 Material. Prefaced concrete masonry units shall consist of concrete masonry units in accordance with ASTM C 90, C 129, and C 145, as applicable with the exposed-to-view-in-place surfaces covered at the point of manufacture with a resin and inert filler or cement and inert filler to produce a smooth tile-like facing.

3.2.2 Surface. The coated surface shall be free from chips, crazes, cracks, blisters, crawling, holes and other imperfections detracting from the appearance of the unit (see 4.3.2).

3.2.3 Dimensional tolerances.

3.2.3.1 Face dimension. The total variation in the finished face dimensions of units shall not be more than 1/16 inch between the largest and smallest unit in any lot of each size.

3.2.3.2 Bed-depth dimension. The total variation in the bed depth (through wall) dimension of units shall be not more than plus or minus 1/8 inch from the specified size.

3.2.3.3 Distortion. The distortion of the plane and edges of the face of individual units from the corresponding plane surface and edges of the concrete masonry unit shall not exceed 1/16 inch.

3.2.4 Facing requirements.

3.2.4.1 Resistance to crazing. There shall be no crazing, cracking, or spalling of facing material when the unit is tested in accordance with 4.4.2.1.

3.2.4.2 Resistance to chemicals. The condition of the surface shall show no change after testing with each of the following chemicals for the duration in 4.4.2.2, except that a slight etching will be allowed in the sodium hydroxide and a slight discoloration will be allowed in the chlorine bleach test.

Undiluted ammonium hydroxide.
Glacial acetic acid.
Citric acid (20 percent).
Sulfuric acid (10 percent.
Tannic acid (4 percent).
Sodium hydroxide (10 percent).
Trisodium phosphate (5 percent).
Hydrogen peroxide (3 percent).
Chlorine bleach (1 percent).
Household detergent, Tide (10 percent).
Vegetable oil.
Blue-black ink (Waterman's).
Tincture of iodine (2 percent).
Ethyl alcohol, industrial denatured (95 percent).

3.2.4.3 Adhesion. When viewed without magnification, there shall be no failure of the adhesion of the facing material to the to the concrete-masonry-unit material at the time of failure of the entire pre-faced unit (see 4.4.2.3).

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3.2.4.4 Water absorption. Increase in weight from water absorption shall not exceed 0.5 percent (see 4.4.2.4).

3.2.4.5 Abrasion. The wear factor shall not exceed 130 when tested in accordance with 4.4.2.5.

3.2.4.6 Surface burning characteristics. Prefaced concrete masonry units shall be noncombustible. When tested in accordance with 4.4.2.6 the surfaces of the prefaced units, including any surfaces of the units exposed by cutting, shall have a flame spread rating not higher than 25 without evidence of continued progressive combustion and shall have a smoke developed rating not higher than 50.

3.2.4.7 Color and color change. Color, tint or texture shall be as specified (see 6.2). When tested in accordance with 4.4.2.7 and then compared with an unexposed unit there shall be no change in color, gloss, or texture.

3.2.4.8 Soiling and cleansability. There shall be not more than a trace of the stain visible when tested in accordance with 4.4.2.8.1 and 4.4.2.8.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all the masonry units of the same type, grade, class, form and style offered for delivery at the same time.

4.2.2 Visual and dimensional examination of end item. Sampling for these characteristics shall be in accordance with level S-3 in MIL-STD-105.

4.2.3 Sampling for tests. Samples for test shall be in accordance with ASTM C 140.

4.3 Examination.

4.3.1 Visual and dimensional. Mansonry units selected in accordance with 4.2.2 shall be examined for defects listed in table I. The Acceptable Quality Levels (AQL's) in accordance with MIL-STD-105, shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Form, type, class, and grade not as specified	X	
Finish not as specified	X	
Color and texture not as specified	X	
Dimensions of units not within the specified tolerance	X	
Defects or damage affecting serviceability	X	
Defects or damage not affecting serviceability		X

4.3.2 Visual, prefaced units. View the prefaced surface of block at a distance of 5 feet normal to each block using daylight without direct sunlight (see 3.2.2).

4.3.3 Inspection of preparation for delivery. The packing and marking of the concrete masonry units shall be inspected to determine conformance with the applicable requirements of section 5. The packing and marking requirements of section 5 identified to referenced specifications and standards shall be inspected in accordance with the applicable requirements of these documents. The packing requirements of section 5 not identified to referenced specifications shall be examined for the defects listed in table II in accordance with MIL-STD-105, using an AQL of 4.0 percent defective.

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TABLE II. Classification of preparation for delivery defects

Examine	Defects
Packing	Cushioning and separators not provided. Fiberboard boxes not waterproof sealed with tape. Case liners not provided. Wood boxes not modified by the addition of skids as specified. Closure and strapping not as specified. Strapping not zinc coated when applicable. Pallets not provided when specified.

4.4 Test methods.

4.4.1 Form A, unglazed. Conduct tests in accordance with ASTM C 140 (see 1.1, 3.1.2, and 3.1.3).

4.4.2 Form B, prefaced.

4.4.2.1 Crazing. Determine resistance to crazing in accordance with ASTM C 426 (see 3.2.4.1).

4.4.2.2 Resistance to chemicals. Apply a convenient quantity of the following chemicals one at a time on the facing material and leave uncovered with a watch glass for the duration of the following test periods. For volatile chemicals place a saturated wad under the watch glass and keep saturated:

Chemical	Duration of test, hours
Undiluted ammonium hydroxide	12
Glacial acetic acid	24
Citric acid (20 percent)	24
Sulfuric acid (10 percent)	24
Tannic acid (4 percent)	24
Sodium hydroxide (10 percent)	24
Trisodium phosphate (5 percent)	24
Hydrogen peroxide	24
Chlorine bleach (1 percent)	24
Household detergent, Tide (10 percent)	24
Vegetable oil	24
Blue-black ink (Waterman's)	1
Tincture of iodine (2 percent)	1
Ethyl alcohol, industrial denatured, 95 percent	3

Upon completion of the test cycle wipe the test area clean with a clean cloth (see 3.2.4.2).

4.4.2.3 Adhesion of facing. Place uncapped 4 by 8 by 16 inch prefaced units between steel bearing plates with the upper plate bearing having a spherical head, and with the longest dimension of the masonry unit in the horizontal plane with the cores vertical. Apply the compressive load until failure of the entire unit occurs (see 3.2.4.3).

4.4.2.4 Water absorption. Cut a 2 inch square specimen from each block. Place specimens in an oven maintained at 200° ± 10°F. until the weight of each specimen becomes constant. Remove from oven and air cool to room temperature. Immerse specimens in water at 70° ± 2°F. for 24 consecutive hours. Remove and dry samples superficially with a clean towel to remove surface water only, and reweigh. Determine weight to the nearest 1/10 gram (see 3.2.4.4).

4.4.2.5 Abrasion. Determine the wear factor in accordance with method 6192 in Fed. Test Method Std. No. 141 using Taber Abraser Research Model 503 with a Taber CS-17 abrasive wheel and a 1,000 gram weight for 500 wear cycles (see 3.2.4.5 and 6.5).

4.4.2.6 Surface burning. Determine flame spread and smoke ratings in accordance with ASTM E 84 (see 3.2.4.6).

4.4.2.7 Color change. Determine change in color in accordance with method 6152 in Fed. Test Method Std. No. 141 for a duration of 500 hours (see 3.2.4.7).

4.4.2.8 Soiling and cleansability.

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4.4.2.8.1 Staining media. Pour a stain consisting of 0.5 gram of oil soluble-dye (C.I. Solvent Orange 7, C.I. 12140), 1.0 gram of lanolin, and 5.0 grams of S. A. E.10 oil one at a time on an area approximately 1/2 inch diameter on the preface side of units. Allow the stain to remain on the specimens for a period of four days. Clean the specimen in accordance with method 6142 in Fed. Test Method Std. No. 141 using Lestoil, industrial grade, as a cleaning agent. Examine the cleaned surface for the following degrees of intensity (see 3.2.4.8):

(a) Stain completely removed.
(b) Trace.
(c) Evident.
(d) Pronounced.
(e) Very pronounced.

4.4.2.8.2 Spotting media. Apply each of the following stains on each specimen:

(a) Writing pencil, number 2. Apply the pencil soil by wedging it in a glass tube that fits snugly but floats freely in a wooden block. Apply the soil by placing a 500-gram weight on top of the pencil in such a manner that the soil is uniform in width and thickness.

(b) Magic marker. Apply the magic marker soil by wedging it in a glass tube that fits snugly but floats freely in a wooden block. Apply the soil by placing a 500-gram weight on top of the magic marker in such a manner that the soil is uniform in width and thickness.

(c) Lanolin. Smear the lanolin on the surface using an artist's or similar brush.

(d) Carbon paper. Apply the carbon paper soil by placing a 500-gram weight on1 square inch of the paper and drawing the weighted paper across the surface in a single stroke.

Clean spotting media (a), (c), and (d), using the same procedures in 4.4.2.8.1 until each is removed. Clean spotting media (b) using paint thinner conforming to TT-T-295 (see 3.2.4.8).

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unglazed units. Unglazed units shall be palletized in accordance with the applicable requirements of MIL-STD-147.

5.1.1.2 Prefaced units. Prefaced units shall be packed in fiberboard, triple wall fiberboard, nailed and lock-corner wood or cleated plywood boxes conforming to PPP-B-636 (class weather-resistant); PPP-B-640 (class 2, grade A); PPP-B-621 (class 2); and PPP-B-601 (overseas type), respectively. Separators, conforming to PPP-F-320 (type CF,domestic class, grade NT), shall be used to prevent glazed surfaces from abrasion. Cushioning shall be used to fill all voids and insure a compact fit. Fiberboard boxes shall be waterproof sealed with tape in accordance with the appendix to the applicable box specification. Wood boxes shall be lined with sealed case liners conforming to MIL-L-10547, and the appendix thereto. Wood boxes exceeding 250 pounds gross weight shall be modified by the addition of skids in accordance with the applicable box specification. Boxes shall be closed and strapped in accordance with the applicable box specification except that flat steel strapping shall be type I, class B.

5.1.2 Level B. The units shall be packed as specified for level A except that boxes shall be domestic type or class, as applicable, and waterproof sealing, case liner and flat steel strapping requirements do not apply.

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5.1.3 Level C. The units shall be packed in a manner to insure arrival at destination in satisfactory condition and be acceptable to carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules, or National Motor Freight Classification Rules.

5.1.4 Palletization. When specified (see 6.2), the packed glazed units shall be palletized in accordance with MIL-STD-147 for type I load.

5.2 Marking (see 6.2).

5.2.1 Civil agencies. In addition to any special markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Form A, load bearing.

6.1.1.1 Grade U. This grade of masonry units are intended for use in unprotected exterior walls below grade and also for unprotected exterior walls above grade which may be exposed to frost action.

6.1.1.2 Grade P. This grade or masonry units are intended for use in protected exterior walls below grade and also for protected walls above grade that may be exposed to frost action.

6.1.1.3 Grade G. These grade of masonry units are intended for general use above grade in walls not subjected to frost action, or in walls protected from weather.

6.1.2 Form A, nonload bearing. Hollow nonload bearing units are intended for nonload bearing partitions.

6.1.3 Form B. Prefaced masonry units are intended for use wherever a masonry wall with a glazed surface is desired.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, class, form, and style (see 1.2).
(c) Styles a, b, c, d, e, g, and i, specify size (see 6.4).
(d) Style f, specify radius (1 or 2 inches).
(e) Style h, specify size, if special.
(f) Style j, specify size and shape.
(g) Whether a particular finish, texture, or uniformity of color required on form B units (see 3.2.4.7 and 6.3).
(h) Size of lot, if different from 4.2.1.
(i) Selection of applicable level of packing required (see 5.1).
(j) If palletizing is required (see 5.1.4).
(k) If marking is required (see 5.2).

6.3 Color of form B. To insure continuity of color in prefaced units for interior use, all units for a given project should be purchased from one producer. Exterior use of prefaced units shall take into account that solar radiation changes the color of these surfaces with continued exposure.

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6.4 Available sizes. Concrete units are available in 8 by 16, 8 by 8, and 4 by 16 dimensions or nominal wall thicknesses of 12, 8, 6 and 4 inches. Two inch blocks are available in these face dimensions for special applications. Concrete units of other face or wall thickness, and other shapes are also available for special applications.

6.5 Abrasive unit. The abrasive equipment and abrasive wheel in 4.4.2.5 can be obtained from:

Taber Instrument Corporation
107 Goundry Street
North Tonowanda, New York 14150

6.6 Spotting media. Magic marker model 67 referenced in 4.4.2.8.2 can be obtained from:

Speedy Chemical Company
84-00 73rd Avenue
Glendale, New York 11227

MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - None

Review activities:

Army - ME
Navy - YD

User activity:

Navy - CG

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